1.5 Infinite Descent

WORKSHEET

The question

STEP III 2022, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself In part (i), look at $a^3 + 2b^3 + 4c^3 = 0$. Two of the three terms are obviously even. What does that tell you about the third?

STAGE 1 The first descent

Ask yourself The new equation is $4p^3 + b^3 + 2c^3 = 0$. It has the same coefficients, just rotated. Can you do the same trick again?

Pause and try Turn this into a proof that no non-zero solution exists. What, exactly, should you assume is the "smallest"?

STAGE 2 A different prime

Ask yourself In $9a^3 + 10b^3 + 6c^3 = 0$, parity doesn't obviously help. Look at the coefficients 9, 10, 6. Which prime divides two of them?

STAGE 3 Squares modulo 3

Ask yourself Every integer is of the form $3n$, $3n+1$ or $3n-1$. What are their squares, modulo 3?

Ask yourself If $a^2 + b^2 = 3c^2$ and $a = 3p$, $b = 3q$, what can you say about c?

STAGE 4 A twist: the equation changes

Ask yourself In $a^2 + b^2 + c^2 = 4abc$ the right-hand side is a multiple of 4. What are the squares modulo 4?

Ask yourself This is *not* the same equation: the 4 has become an 8. Does the descent still work?

What's really going on

Fermat, descent and his Last Theorem

Fermat used infinite descent to prove that $x^4 + y^4 = z^2$ has no solutions in positive integers. This is the only case of his famous "Last Theorem" for which he left a proof, and it settles the case $n = 4$ of $x^n + y^n = z^n$. Euler later used descent, with a gap that was later filled, for $n = 3$. Descent is still the starting point of the modern theory: the proof of the Mordell–Weil theorem about curves like the one in Section 1.9 is a sophisticated descent.

Local obstructions

Every proof in this section worked by looking at remainders modulo some number. If an equation has no solutions modulo n for some n, or no solutions in real numbers, it certainly has no solutions in integers. Such reasons are called *local obstructions*, and at university you will meet the *p-adic numbers*, a way of studying all the remainders modulo $p, p^2, p^3, \ldots$ at once.

A wonderful theorem of Hasse and Minkowski says that for *quadratic* equations like $a^2 + b^2 = 3c^2$, local obstructions are the only obstructions: if there are no "local" reasons to fail, there really is a rational solution. For cubic equations this breaks down spectacularly. Ernst Selmer showed in 1951 that

$$3x^3 + 4y^3 + 5z^3 = 0$$

has solutions modulo every whole number, and in real numbers, but no solution in non-zero integers. Explaining when and why the "local-to-global" principle fails is a major theme of modern number theory.

An equation that *does* have solutions

Part (iv) has a famous cousin: $x^2 + y^2 + z^2 = 3xyz$, the *Markov equation*. It has infinitely many solutions in positive integers, starting with $(1,1,1)$, $(1,1,2)$, $(1,2,5)$, $(1,5,13)$ and

$(2, 5, 29)$, and they are connected to how well irrational numbers can be approximated by fractions. See exercise 4.

Try this next

1 **A STEP I warm-up.** *(STEP I 2010 Question 8.)* Show that the only integer solution of $a^3 + 3b^3 = 9c^3$ is $a = b = c = 0$. Then, by considering final digits, show the same for $p^4 + 2q^4 = 5r^4$.

2 **A modulus with no descent needed.** Show that $x^2 + y^2 = 4z + 3$ has no integer solutions at all, just by working modulo 4.

3 **Rational points.** Use part (iii) to show that there is no point with *rational* coordinates on the circle $x^2 + y^2 = 3$. (Contrast this with $x^2 + y^2 = 2$, which contains $(1, 1)$.)

4 **Markov triples.** Show that if (x, y, z) satisfies $x^2 + y^2 + z^2 = 3xyz$, then so does $(x, y, 3xy - z)$. Starting from $(1, 1, 1)$, generate the triples listed above. (This "jumping" trick is the opposite of descent.)

Further reading

- **Harold Davenport, *The Higher Arithmetic*,** Chapters 2 and 5.
- **Joseph Silverman, *A Friendly Introduction to Number Theory*** (Pearson). Very readable; includes Fermat's descent proof for $x^4 + y^4 = z^2$.
- **Simon Singh, *Fermat's Last Theorem*** (Fourth Estate). The story, from Fermat's margin to Wiles's proof, for a general reader.

The full guided solution is in Section 1.5:
https://beyondstep.keithmcnulty.org/ch01/infinite-descent.html